

Mathematical Statistics Assignment 1

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1.4.3

The data in Table 1.9 are presented to illustrate the role of renewable energy consumption in the U.S. energy supply in 2007 (source: <http://www.eia.doe.gov/fuelrenewable.html>). Renewable energy consists of biomass, geothermal energy, hydroelectric energy, solar energy, and wind energy.

- (a) Construct a bar graph.
- (b) Construct a Pareto chart.
- (c) Construct a pie chart.

TABLE 1.9 Renewable Energy Consumption.	
Source	Percentage
Coal	22%
Natural gas	23%
Nuclear electric power	8%
Petroleum	40%
Renewable energy	7%

Figure 1: Table 1.9: Renewable Energy Consumption

Solution:

- (a) Bar graph

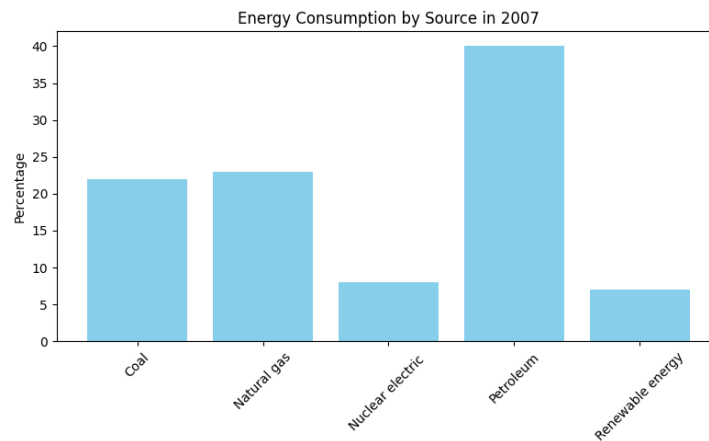


Figure 2: Bar Graph of Energy Consumption

- (b) Pareto chart

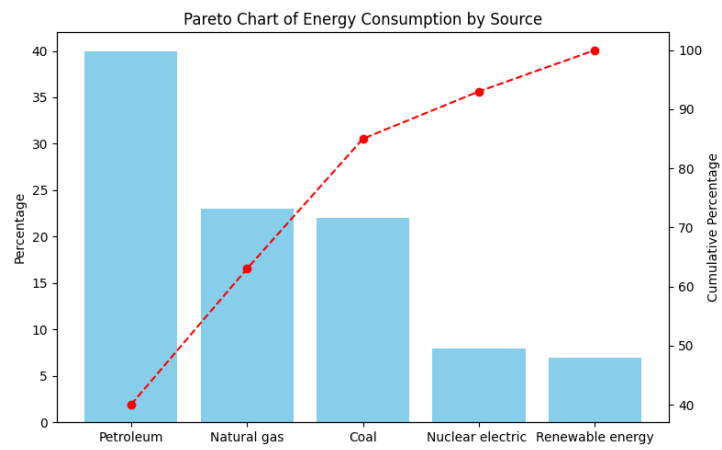


Figure 3: Pareto Chart

(c) Pie chart

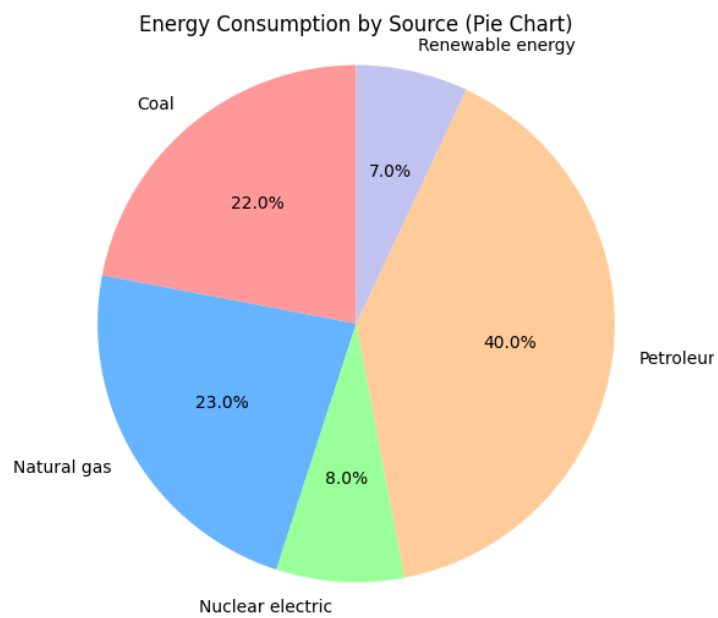


Figure 4: Pie Chart

1.4.14

The following table gives radon concentrations in pCi/liter (picocurie per liter) obtained from 40 houses in a certain area.

2.9	0.6	13.5	17.1	2.8	3.8	16.0	2.1	6.4	17.2
7.9	0.5	13.7	11.5	2.9	3.6	6.1	8.8	2.2	9.4
15.9	8.8	9.8	11.5	12.3	3.7	8.9	13.0	7.9	11.7
6.2	6.9	12.8	13.7	2.7	3.5	8.3	15.9	5.1	6.0

- Construct a stem-and-leaf display.
- Construct a frequency histogram and interpret.
- Construct a pie chart and interpret.

Solution:

- Stem and leaf

```
"D:\Software Application\Python2125\python.exe" "D:\Masters NPV Semester\3rd Semester\Statistics\Homework\Submit\Homework 1\code1414.py"
Stem-and-Leaf Display:
0 | 5 6
2 | 1 2 7 8 9 9
3 | 5 6 7 8
5 | 1
6 | 0 1 2 4 9
7 | 9 9
8 | 3 8 8 9
9 | 4 8
11 | 5 5 7
12 | 3 8
13 | 0 5 6 6
15 | 9 9
16 | 0
17 | 1 2
Process finished with exit code 0
```

stem	leaves
0	5 6
2	1 2 7 7 9 9
3	5 6 7 7
5	0
6	0 0 2 4 9
7	9 9
8	3 8 8 9
9	4 8
11	5 5 6
12	3 8
13	0 5 6 6
15	9 9
16	0
17	1 1

(b) Histogram and Interpret

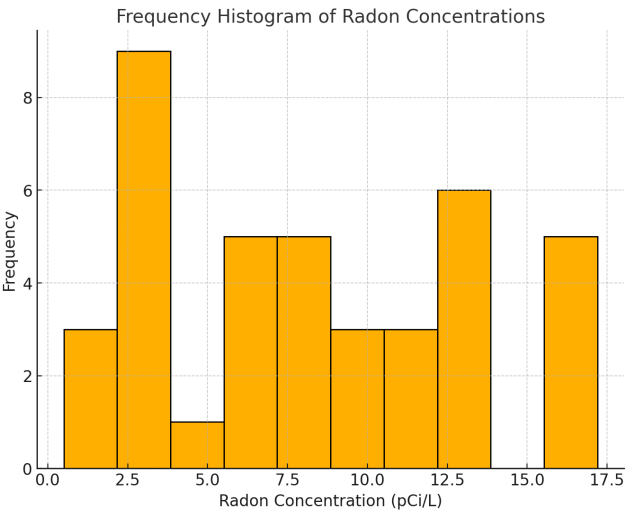
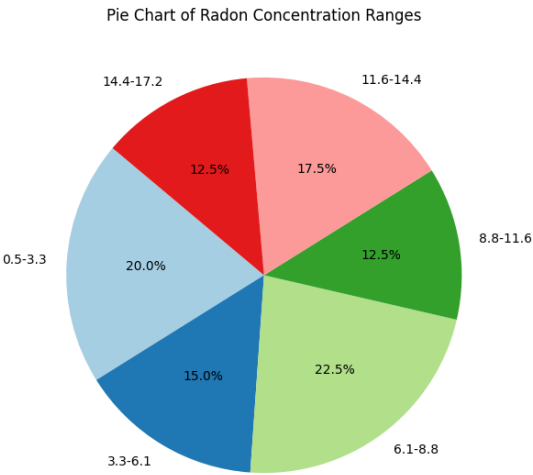


Figure 5: Bar Graph of Energy Consumption

(c) Pie chart and Interpret



1.5.5

Maximal static inspiratory pressure ($PI_{\max}$) is an index of respiratory muscle strength. The following data show the measure of $PI_{\max}$ (cm H₂O) for 15 cystic fibrosis patients.

105	80	115	95	100	85	90	70
135	105	45	115	40	115	95	

- Find the lower and upper quartiles, median, and interquartile range. Check for any outliers and interpret.
- Construct a box plot and interpret.
- Are there any outliers?

Solution:

(a) **Solution:**

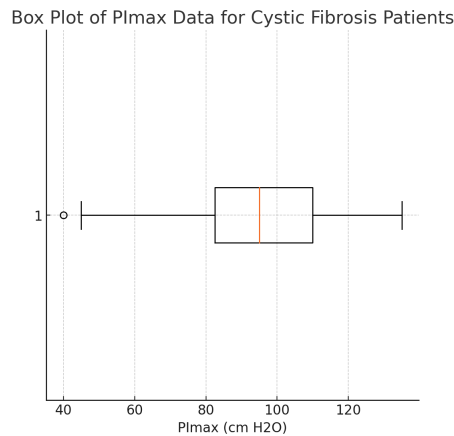
- Lower Quartile ($Q1$) = 80
- Median ($Q2$) = 95
- Upper Quartile ($Q3$) = 115
- Interquartile Range (IQR) = $Q3 - Q1 = 115 - 80 = 35$
- Outlier Boundaries:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR = 80 - 1.5 \times 35 = 27.5$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR = 115 + 1.5 \times 35 = 167.5$$

Since all data points fall between 27.5 and 167.5, **there are no outliers.**

(b) **Box Plot:**



- (c) **Outliers:** Based on the lower and upper bounds for outliers (27.5 and 167.5), there are no outliers in this dataset.

1.5.7

- (a) For any grouped data with l classes with group frequencies f_i and class midpoints m_i , show that

$$\sum_{i=1}^l f_i(m_i - \bar{x}) = 0.$$

- (b) Verify this result for the data given in *Exercise 1.5.6*.

TABLE 1.21 Class and Frequency.					
Class	0–4	5–9	10–14	15–19	20–24
Frequency	5	14	15	10	6

Figure 6: Exercise 1.5.6-TABLE 1.21 Class and Frequency.

Solution:

Solution:

- (a) For any grouped data with l classes and group frequencies f_i and class midpoints m_i , we need to show that:

$$\sum_{i=1}^l f_i(m_i - \bar{x}) = 0.$$

The formula for the mean $\bar{x}$ of grouped data is given by:

$$\bar{x} = \frac{\sum_{i=1}^l f_i m_i}{\sum_{i=1}^l f_i}.$$

We begin by expanding the left-hand side:

$$\sum_{i=1}^l f_i(m_i - \bar{x}) = \sum_{i=1}^l f_i m_i - \sum_{i=1}^l f_i \bar{x}.$$

Since $\bar{x}$ is a constant, we can factor it out from the summation:

$$\sum_{i=1}^l f_i \bar{x} = \bar{x} \sum_{i=1}^l f_i.$$

Now substitute the expression for $\bar{x}$:

$$\bar{x} = \frac{\sum_{i=1}^l f_i m_i}{\sum_{i=1}^l f_i}.$$

Thus:

$$\sum_{i=1}^l f_i \bar{x} = \frac{\sum_{i=1}^l f_i m_i}{\sum_{i=1}^l f_i} \cdot \sum_{i=1}^l f_i = \sum_{i=1}^l f_i m_i.$$

Now substitute back into the original equation:

$$\sum_{i=1}^l f_i (m_i - \bar{x}) = \sum_{i=1}^l f_i m_i - \sum_{i=1}^l f_i m_i = 0.$$

Thus, we have shown that:

$$\sum_{i=1}^l f_i (m_i - \bar{x}) = 0.$$

(b) Verify the result for the data in Exercise 1.5.6.

The data provided in Exercise 1.5.6 is:

Class	Frequency
0 – 4	5
5 – 9	14
10 – 14	15
15 – 19	10
20 – 24	6

First, calculate the midpoints m_i for each class:

$$m_1 = \frac{0+4}{2} = 2, \quad m_2 = \frac{5+9}{2} = 7, \quad m_3 = \frac{10+14}{2} = 12, \\ m_4 = \frac{15+19}{2} = 17, \quad m_5 = \frac{20+24}{2} = 22.$$

Next, calculate the mean $\bar{x}$ of the data:

$$\bar{x} = \frac{\sum_{i=1}^5 f_i m_i}{\sum_{i=1}^5 f_i} = \frac{5(2) + 14(7) + 15(12) + 10(17) + 6(22)}{5 + 14 + 15 + 10 + 6}.$$

First, compute the numerator:

$$5(2) + 14(7) + 15(12) + 10(17) + 6(22) = 10 + 98 + 180 + 170 + 132 = 590.$$

Now, compute the denominator:

$$5 + 14 + 15 + 10 + 6 = 50.$$

Thus, the mean $\bar{x}$ is:

$$\bar{x} = \frac{590}{50} = 11.8.$$

Now, verify the result $\sum_{i=1}^5 f_i(m_i - \bar{x}) = 0$ by calculating each term:

$$\begin{aligned}f_1(m_1 - \bar{x}) &= 5(2 - 11.8) = 5(-9.8) = -49, \\f_2(m_2 - \bar{x}) &= 14(7 - 11.8) = 14(-4.8) = -67.2, \\f_3(m_3 - \bar{x}) &= 15(12 - 11.8) = 15(0.2) = 3, \\f_4(m_4 - \bar{x}) &= 10(17 - 11.8) = 10(5.2) = 52, \\f_5(m_5 - \bar{x}) &= 6(22 - 11.8) = 6(10.2) = 61.2.\end{aligned}$$

Summing these values:

$$-49 - 67.2 + 3 + 52 + 61.2 = 0.$$

Thus, we have verified that:

$$\sum_{i=1}^5 f_i(m_i - \bar{x}) = 0.$$

1.5.10

The radon concentration (in pCi/liter) data obtained from 40 houses in a certain area are given below.

2.9	0.6	13.5	17.1	2.8	3.8	16.0	2.1	6.4	17.2
7.9	0.5	13.7	11.5	2.9	3.6	6.1	8.8	2.2	9.4
15.9	8.8	9.8	11.5	12.3	3.7	8.9	13.0	7.9	11.7
6.2	6.9	12.8	13.7	2.7	3.5	8.3	15.9	5.1	6.0

- Find the mean, variance, and range for these data and interpret.
- Find lower and upper quartiles, median, and interquartile range. Check for any outliers.
- Construct a box plot and interpret.
- Construct a histogram.
- Locate on your histogram $\bar{x}$, $\bar{x} \pm s$, $\bar{x} \pm 2s$, and $\bar{x} \pm 3s$. Count the data points in each of the intervals $\bar{x} \pm s$, $\bar{x} \pm 2s$, and $\bar{x} \pm 3s$ and compare this with the empirical rule.

Solution:

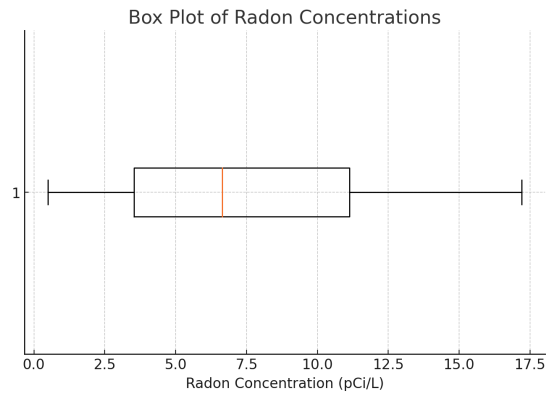
(a) Mean, Variance, and Range

- Mean: $\bar{x} = 7.66$ pCi/L
- Variance: $\sigma^2 = 24.39$
- Range: 16.7 pCi/L

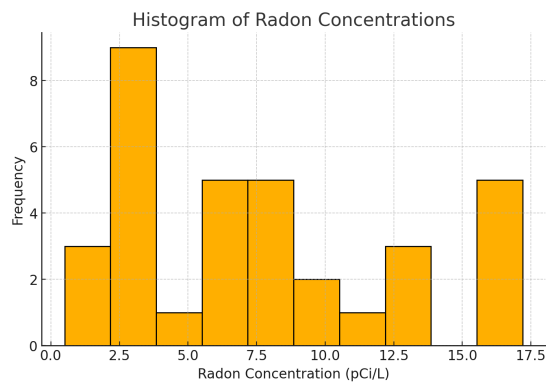
(b) Quartiles, Median, and Interquartile Range (IQR)

- Lower Quartile (Q1): 6.65
- Median: 7.6
- Upper Quartile (Q3): 11.125
- Interquartile Range (IQR): $Q3 - Q1 = 4.475$
- Outliers: No outliers were detected.

(c) Box Plot:



(d) Histogram:



(e) **Empirical Rule Intervals**

- Mean $\pm$ 1 standard deviation: (2.73, 12.60) — 22 data points
- Mean $\pm$ 2 standard deviations: (-2.21, 17.54) — 34 data points
- Mean $\pm$ 3 standard deviations: (-7.15, 22.48) — 34 data points

According to the empirical rule:

- Approximately 68% (22 data points) lie within 1 standard deviation.
- Approximately 95% (34 data points) lie within 2 standard deviations.
- Approximately 99.7% (34 data points) lie within 3 standard deviations.